YAKEEN

NEET 2.0 2027

Physics | Lecture - 7

Basic Maths and Calculus



By- Manish Raj (MR Sir)

Todays Goal

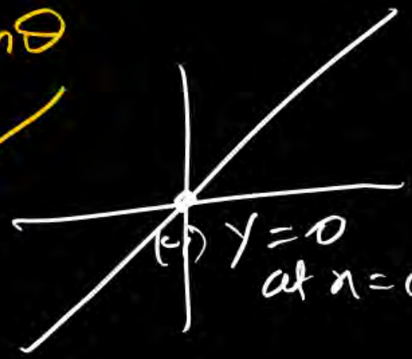
slope of graph & Quadratic equation

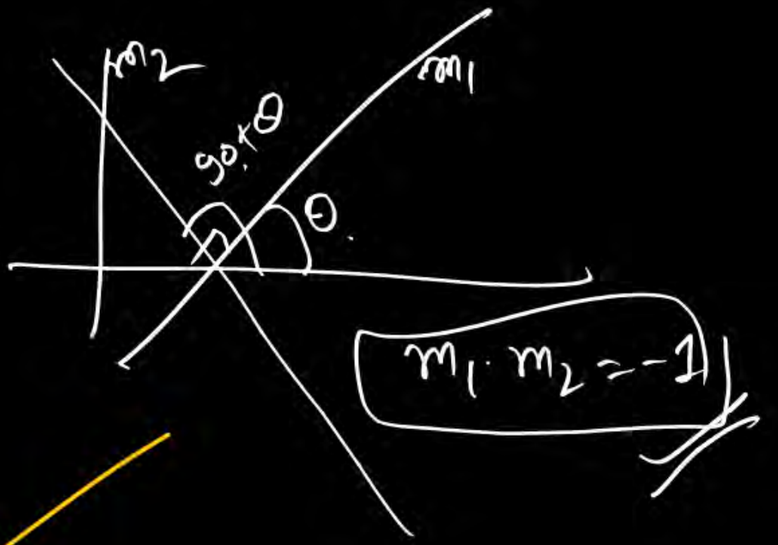
Revision

$y = mx + c$

$x = 0$

$y = c$ $\quad m = \tan\theta$

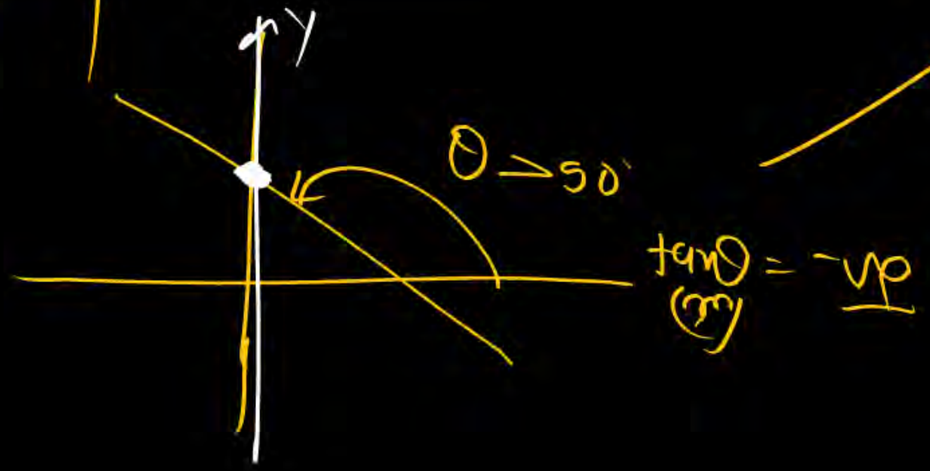


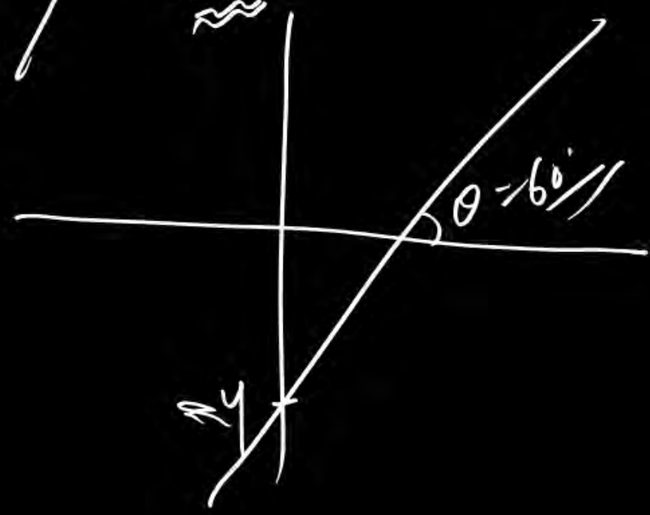


$\theta < 90 \rightarrow m = +ve$

$y = -c$

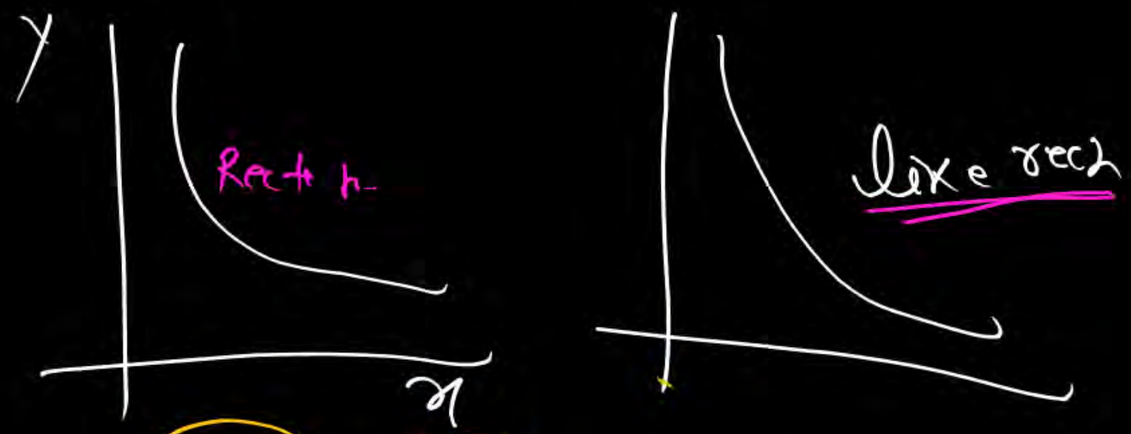
$y = \sqrt{3}x - 4$



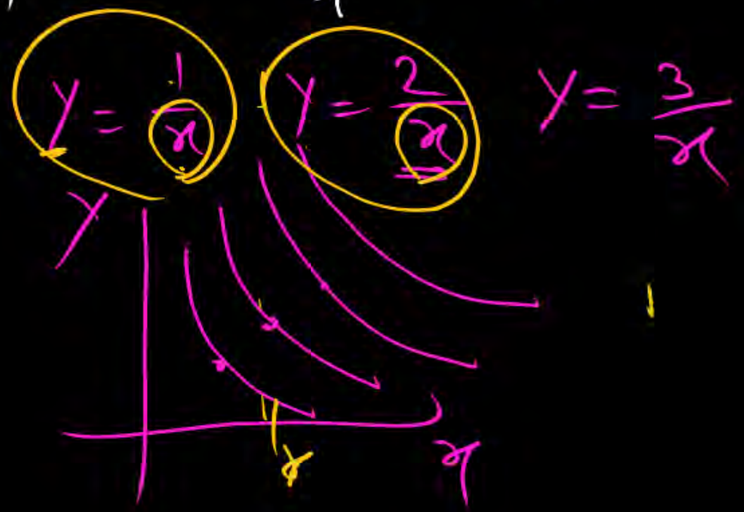


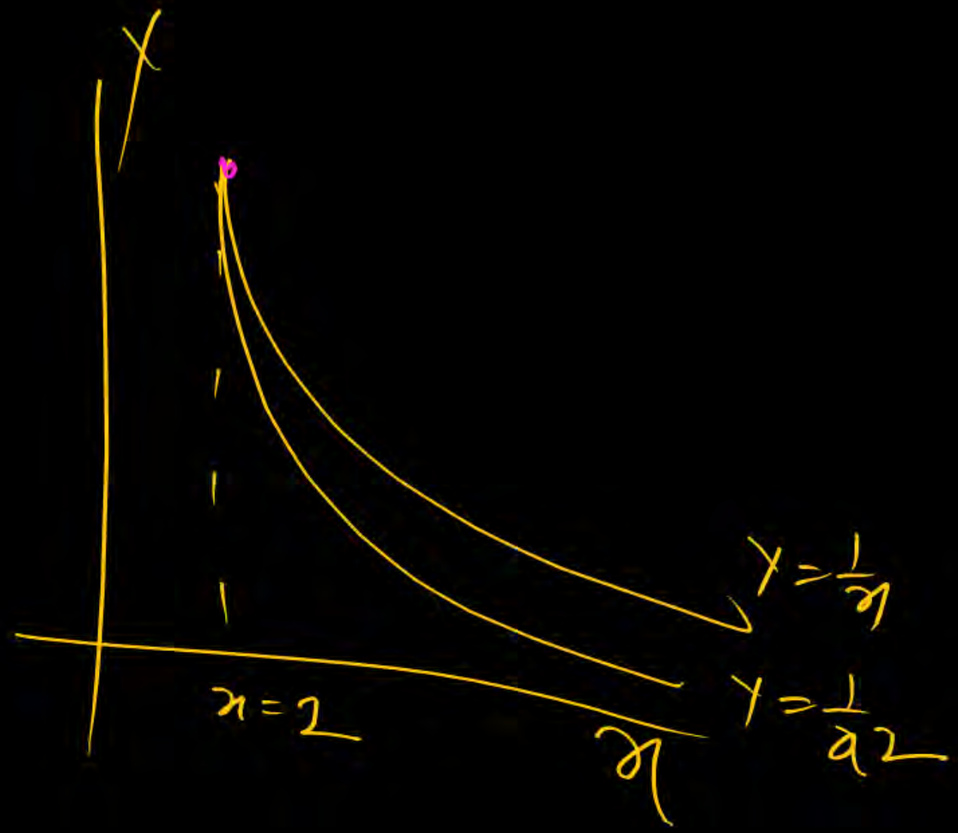
Rectan

$y = \frac{1}{x}$ $y = \frac{1}{x^2}$

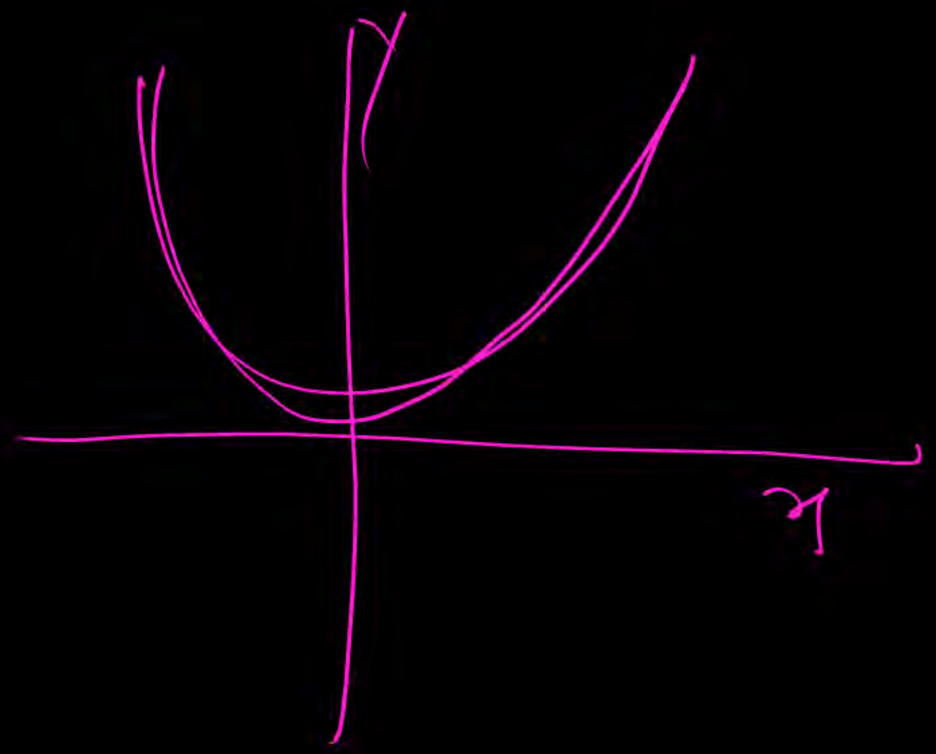


like rec2.





$y = x^2$



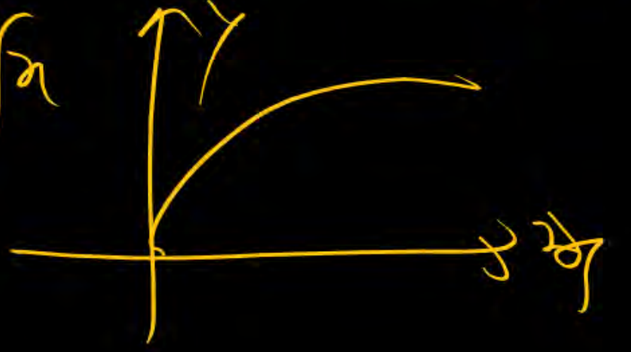
$y = x^2 + 3$



$y = mx + c$

$y = x^2$ → slope नहीं है

$y = \sqrt{x}$

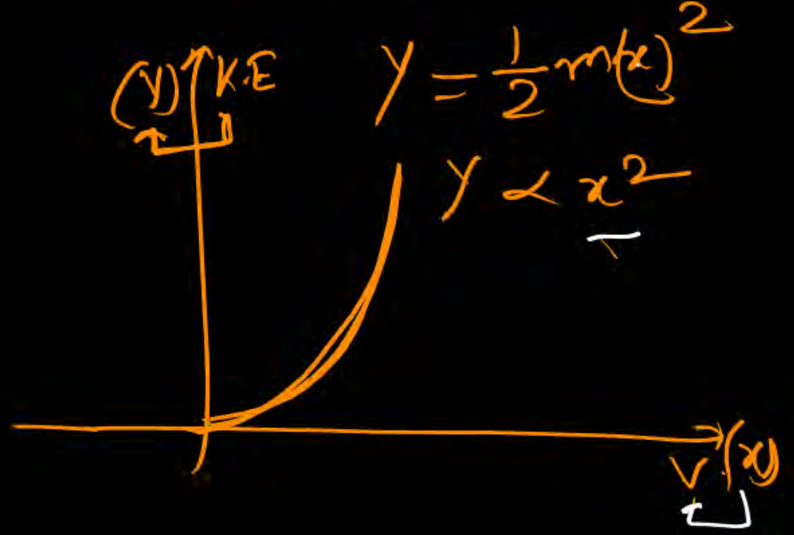


Draw graph b/w K.E and speed and K.E & $(v)^2$

$K.E = \frac{1}{2}mv^2$

step-1 formula

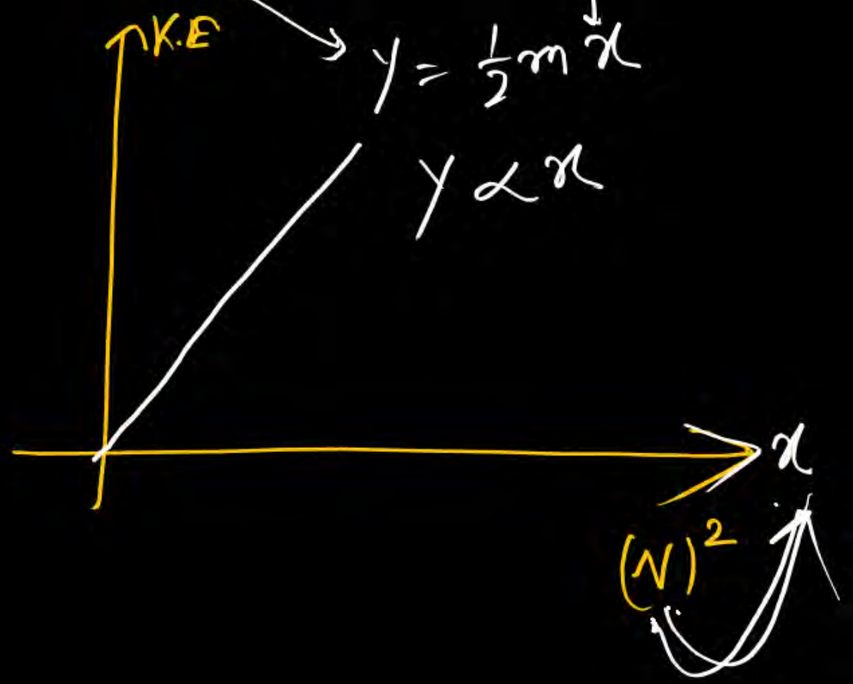
step-2



$y = \frac{1}{2}m(x)^2$

$y \propto x^2$

$KE = \frac{1}{2}m(v)^2$



$y = \frac{1}{2}mx$

$y \propto x$

Copy

MR* Box for graph

* Jis physical quantity ko x-axis any y-axis pe rakha hai, ws ko formula me y and x se replace kar do. then x and y ke relation ke according graph draw karo

H/w

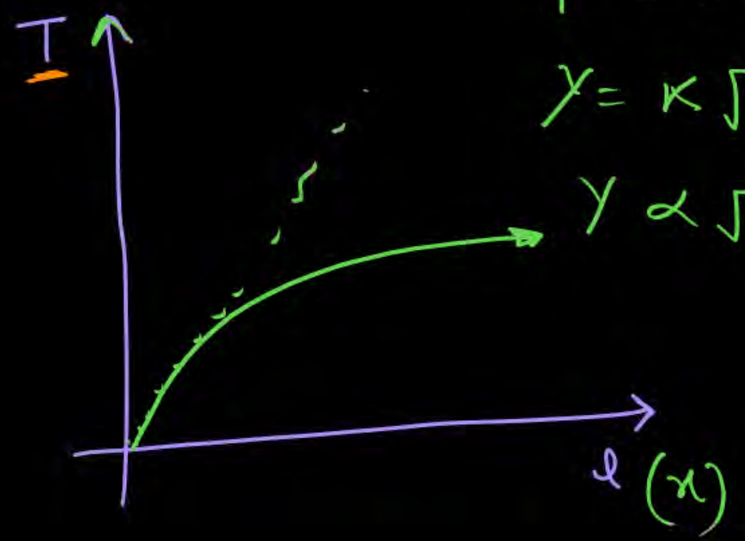
Draw graph b/w Time Period & length of simple pendulum.

$$T = 2\pi\sqrt{\frac{l}{g}}$$ (given in question)

$$T = k\sqrt{l}$$

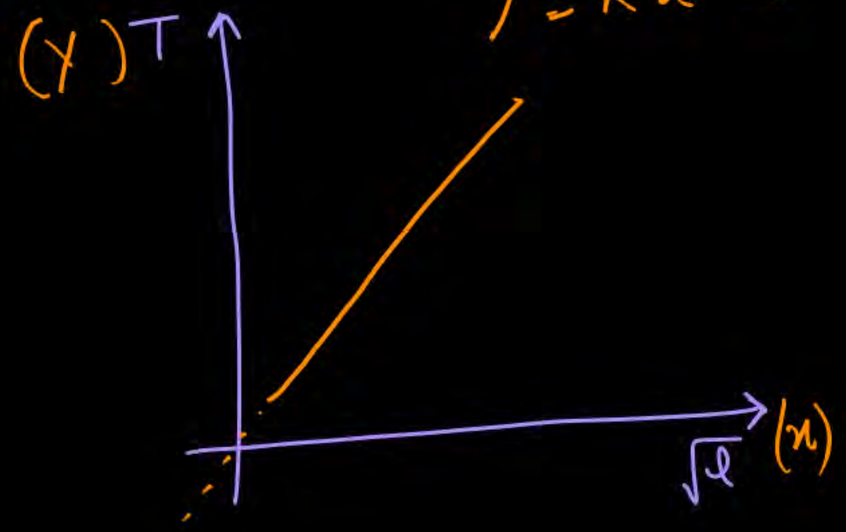
$$Y = k\sqrt{x}$$

$$Y \propto \sqrt{x}$$



$$T = k\sqrt{l}$$

$$Y = kx$$



Kepplers law :- square of time period is propotional to cube of radius. $T^2 \propto R^3$. Draw graph b/w T & R.

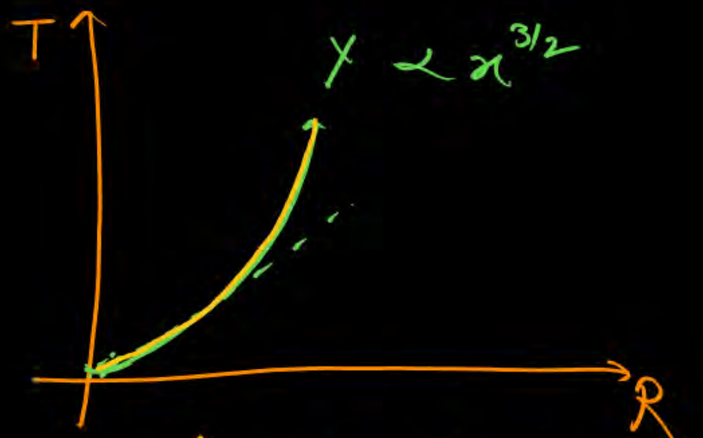
HW

$T^2 \propto R^3$

$T \propto R^{3/2}$

$T \propto R^{1.5}$

$Y \propto x^{3/2}$



like Parabola.

$Y = x^n$

$n=1$ → straight line

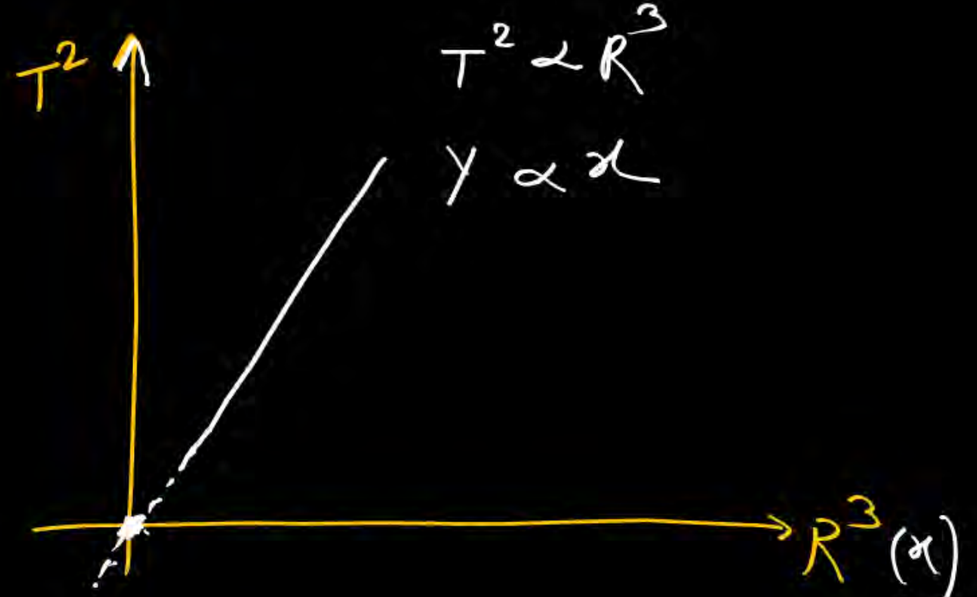
$Y = x^n$ $n=2$ Parabola upward

$Y = \sqrt{x} = x^{1/2}$ $(n=1/2)$



$T^2 \propto R^3$

draw graph b/w T^2 & R^3



Copy

(Q) Draw graph b/w Momentum & $\left(\frac{1}{\text{Velocity}}\right)\frac{1}{V}$; $m = \text{cost}^n$

$P = mv$

$P = \frac{m}{(1/v)}$

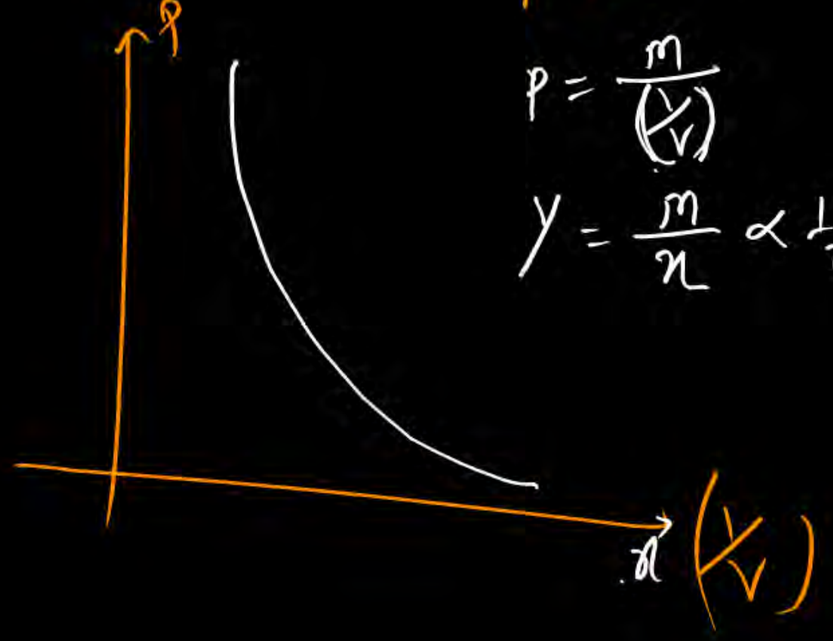
$y = \frac{m}{x} \propto \frac{1}{x}$

$\frac{1}{(1/x)} = x$

$P = mx$

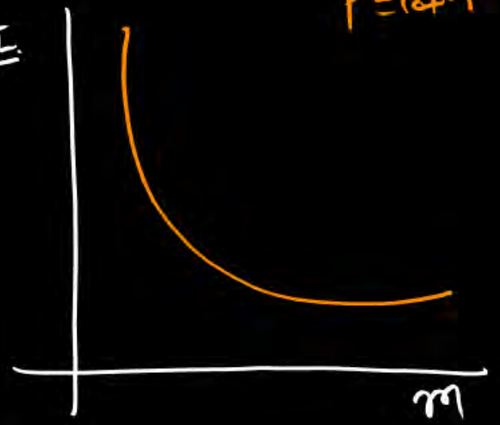
$y = mx$





Q gf $K.E = \frac{p^2}{2m}$ Draw graph b/w K.E & mass

① $p = \text{const}$



② $K.E = \frac{p^2}{2m}$

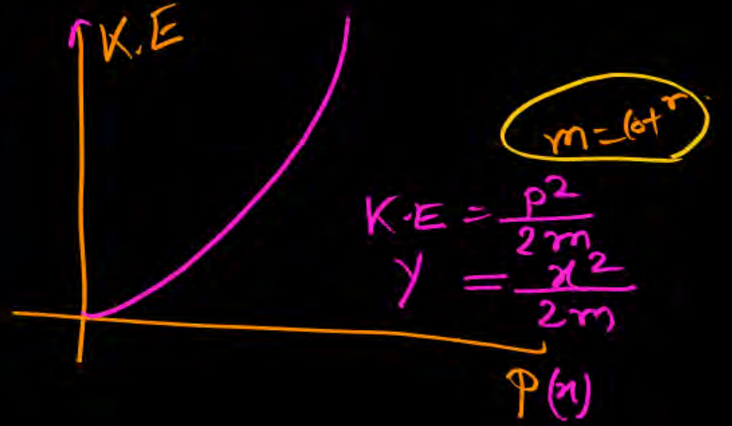
$Y = (x)$



③ $m = \text{const}$

$K.E = \frac{p^2}{2m}$

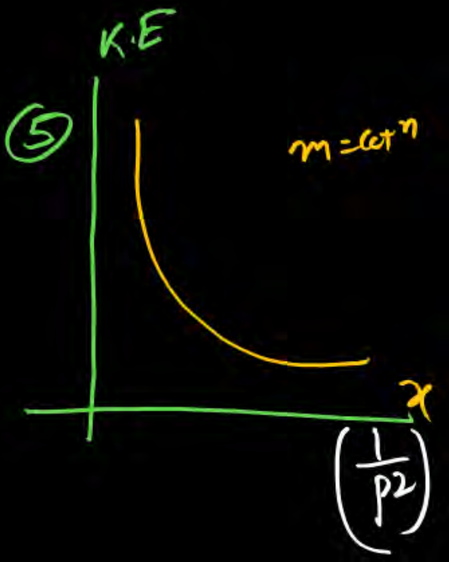
$y = \frac{x^2}{2m}$



④



⑤ $m = \text{const}$



$K.E = \frac{p^2}{2m}$

$y = \frac{1}{\left(\frac{2m}{p^2}\right)} = \frac{1}{2mx}$

$Y \propto \frac{1}{x}$

copy

Equation of circle :-

distance formula.

$$R = \sqrt{(x-x_0)^2 + (y-y_0)^2}$$

$$R^2 = (x-x_0)^2 + (y-y_0)^2$$

eqn of circle

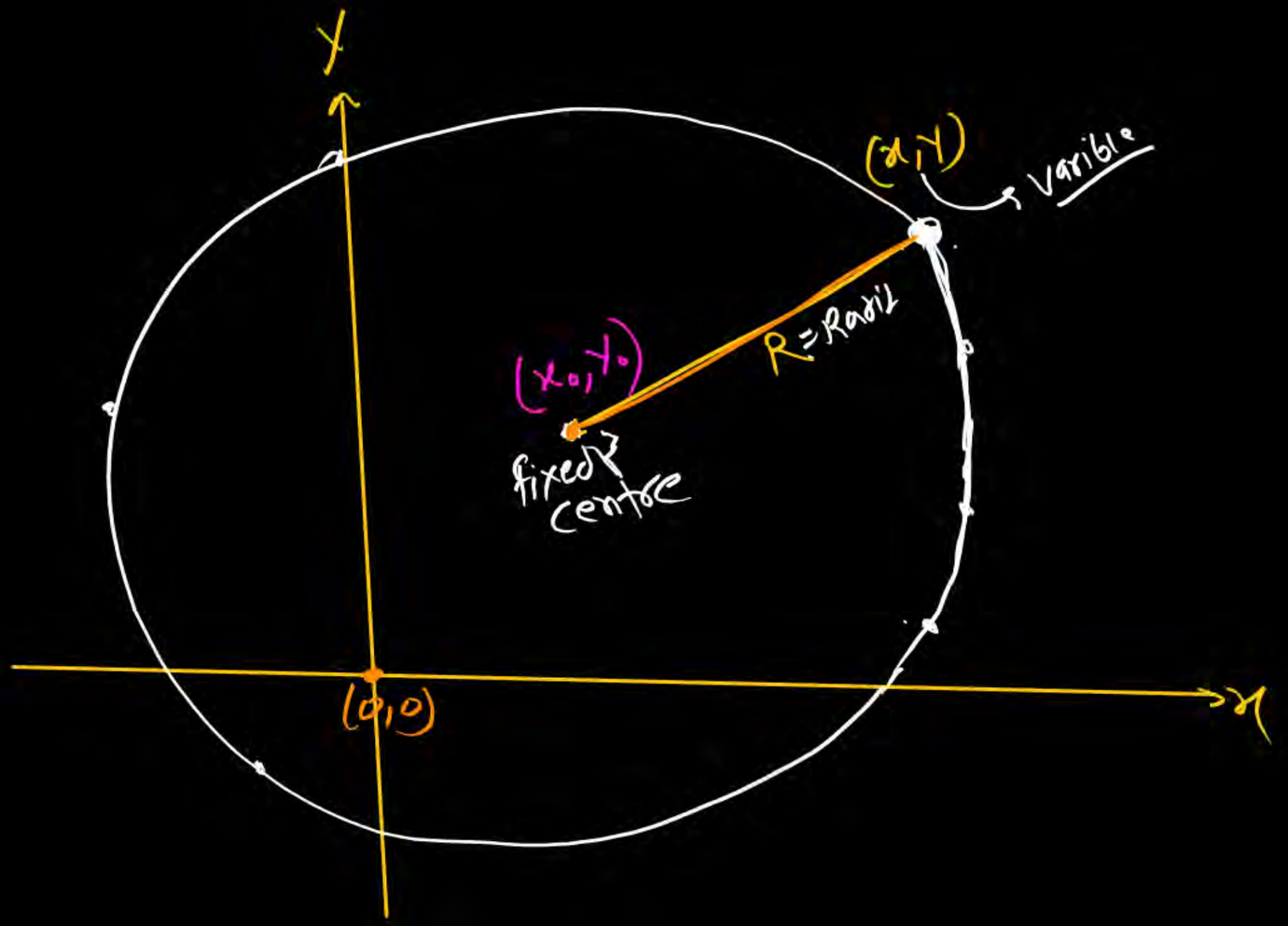
- R = Radius (x_0, y_0) → centre

of centre (x_0, y_0) at origin $(0,0)$

$$R^2 = x^2 + y^2$$

eqn of circle

$$\frac{x^2}{R^2} + \frac{y^2}{R^2} = 1$$



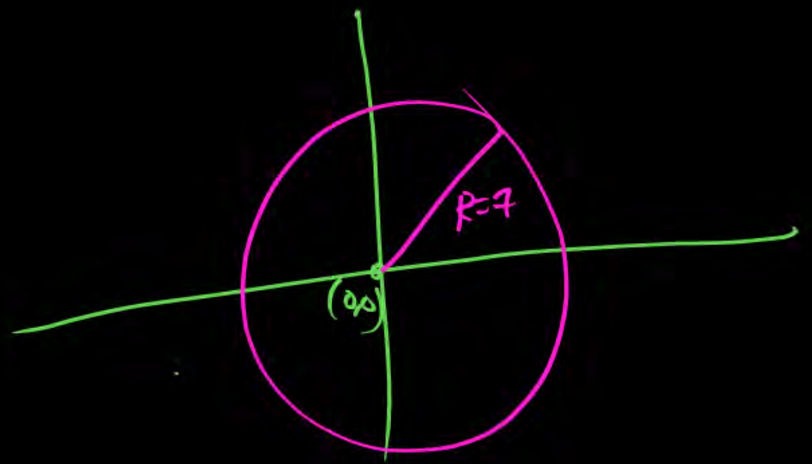
If $x^2+y^2=49$ Draw graph b/w x and y

→ eqn of circle

→ $(x-x_0)^2+(y-y_0)^2=R^2$

$x_0=0$
$y_0=0$

$R^2=49$
$R=7$



Question

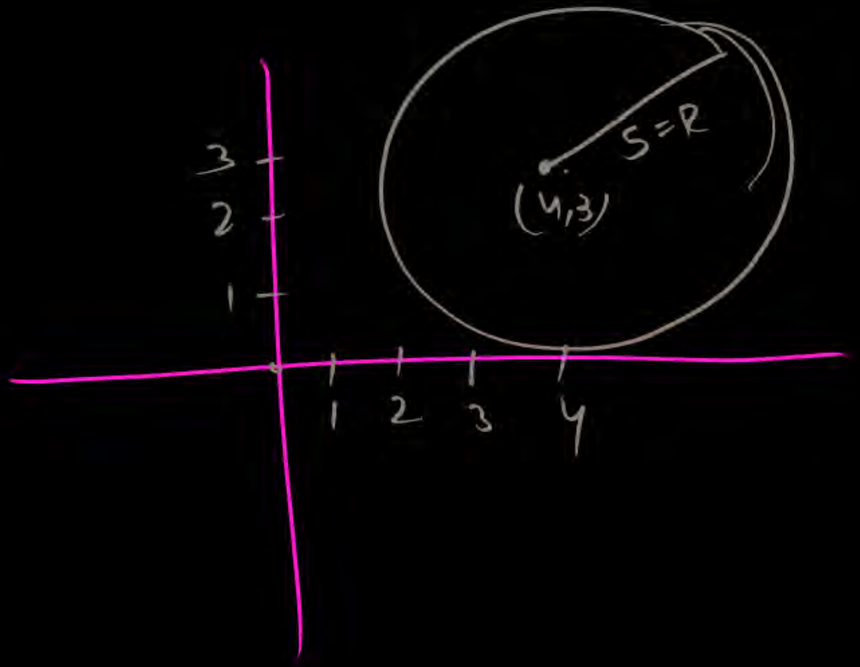
$(x-4)^2 + (y-3)^2 = 25$. Find radius and centre of circle.

$$(x-x_0)^2 + (y-y_0)^2 = R^2$$

Compare

$$\begin{cases} x_0 = 4 \\ y_0 = 3 \end{cases}$$

$R = 5$



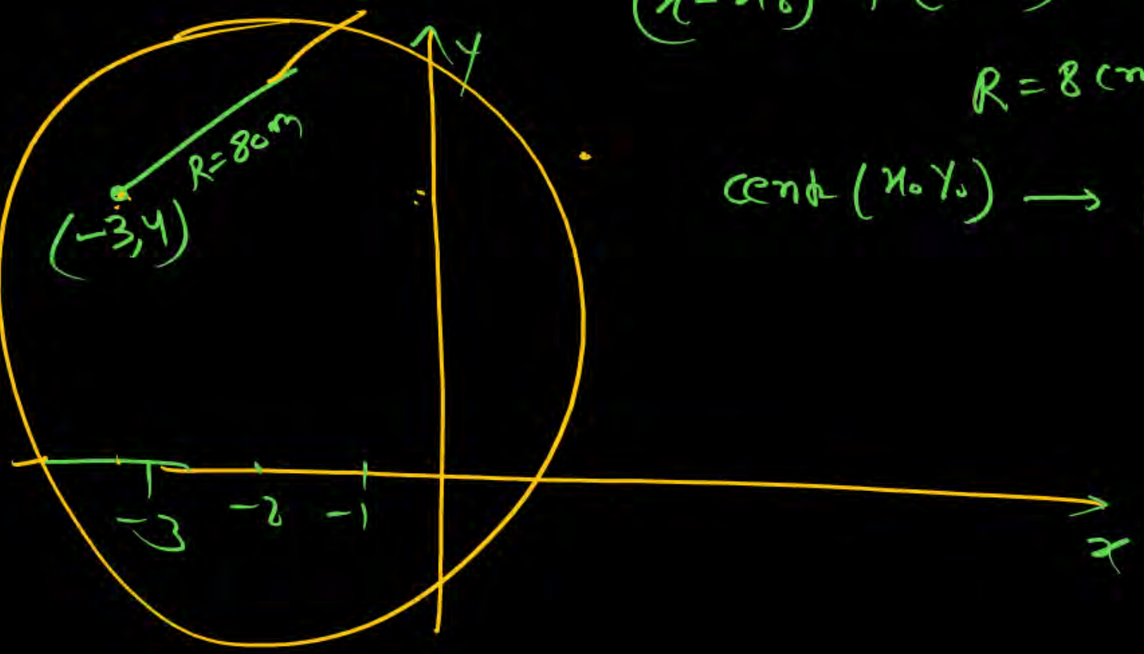
Eqn

$$(x+3)^2 + (y-4)^2 = 64$$

$$(x-x_0)^2 + (y-y_0)^2 = R^2$$

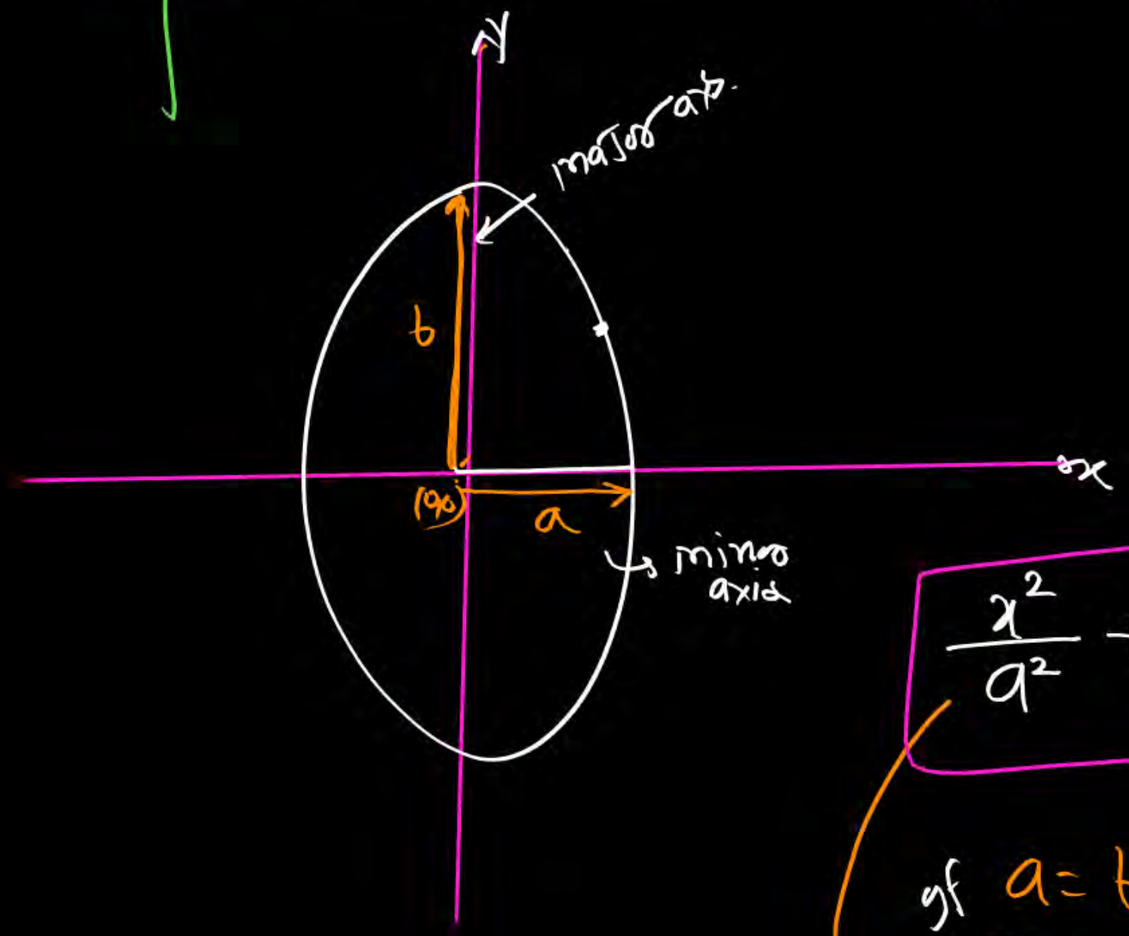
$R = 8\ cm$

centre $(x_0, y_0) \longrightarrow (-3, +4)$





equation of ellipse



eqn of circle

$$x^2 + y^2 = R^2$$

$$\frac{x^2}{R^2} + \frac{y^2}{R^2} = 1$$

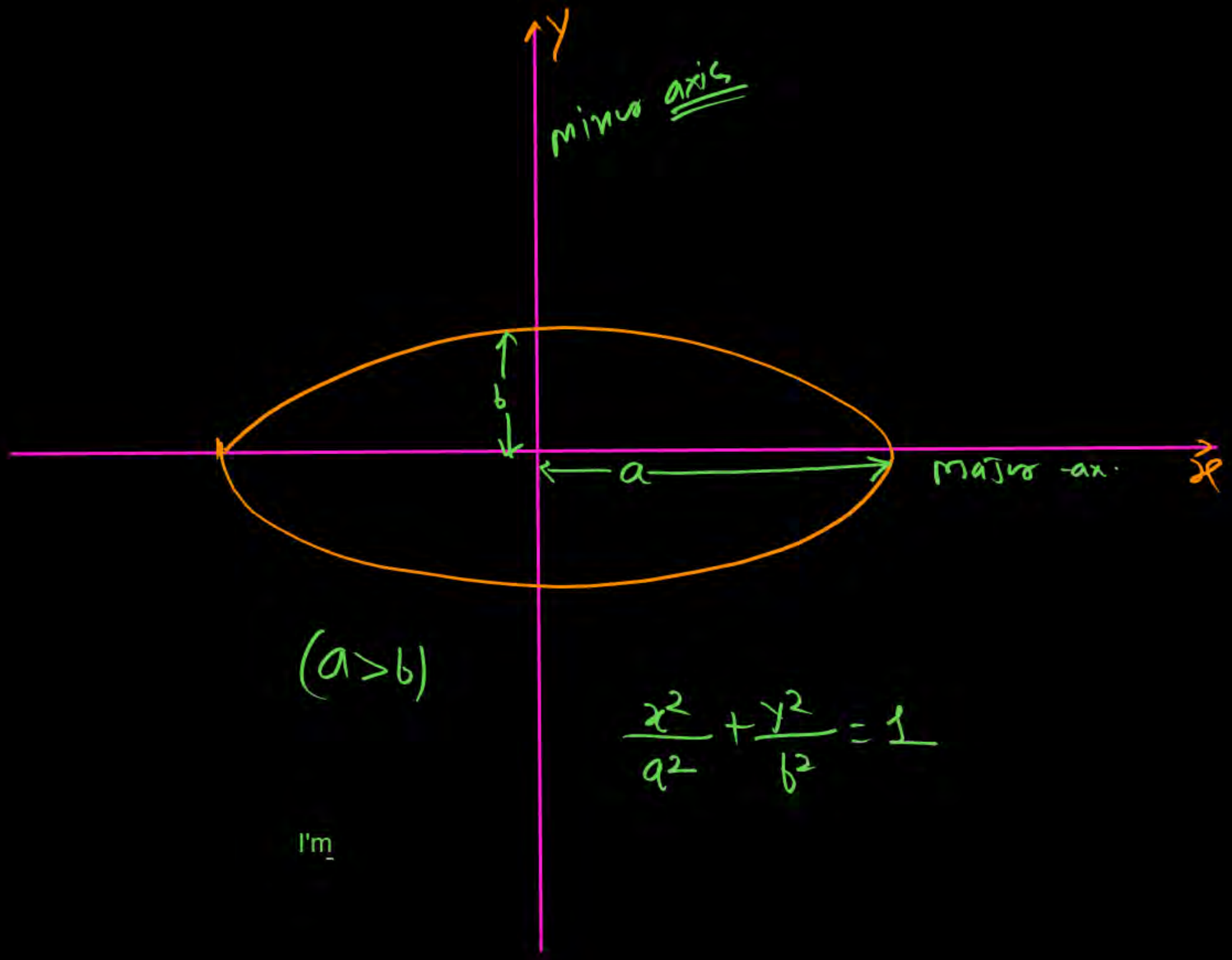
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

ellipse eqn

if $a = b = R$

$$\frac{x^2}{R^2} + \frac{y^2}{R^2} = 1$$

circle



$(a > b)$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

I'm

Question

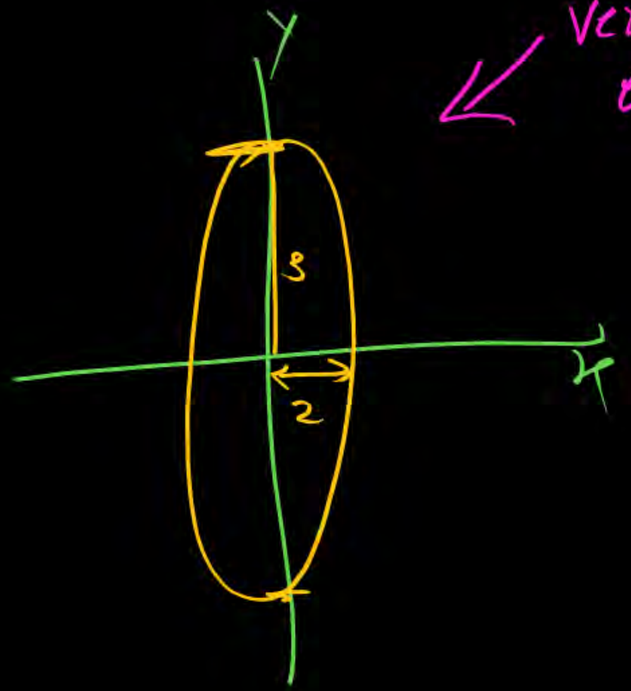
$$\frac{x^2}{4}+\frac{y^2}{9}=1$$

Draw graph between x and y

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

$a^2=4$ $\quad b^2=9$

$a=2$ $\quad b=3$

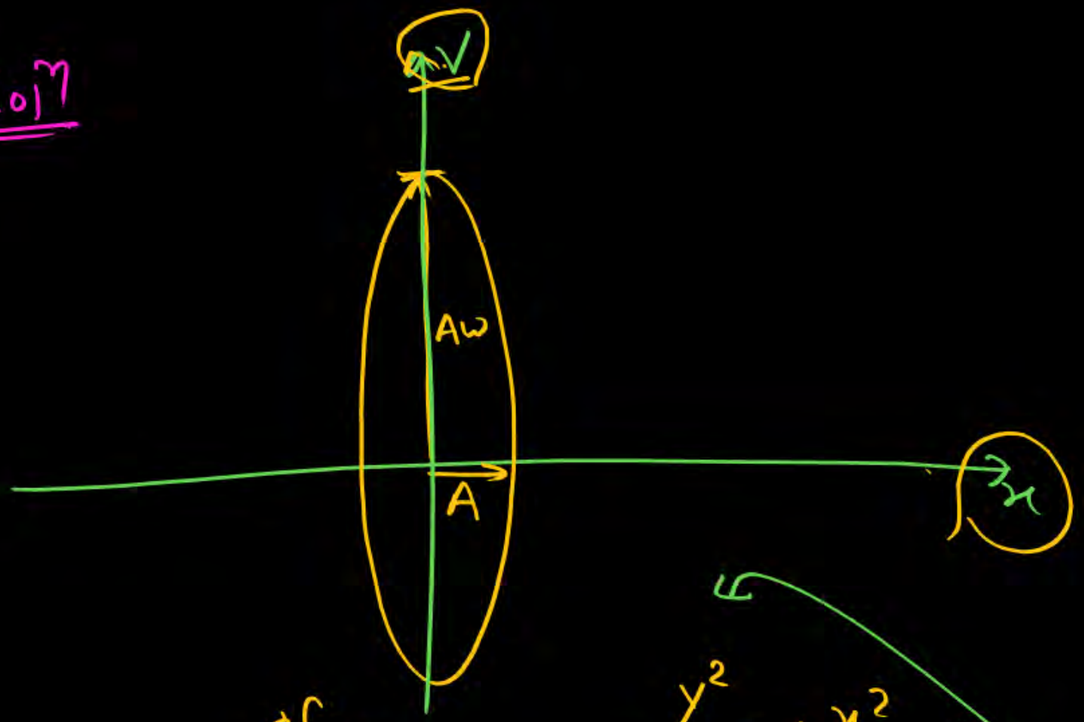


Vertical ellipse

Q. Relation b/w velocity and Position of simple harmonic motion

$V = \omega\sqrt{A^2 - x^2}$ where [ω & A → cost^n] then draw graph between [v and x].

Soln



$$V = \omega\sqrt{A^2 - x^2}$$

$$\frac{V}{\omega} = \sqrt{A^2 - x^2}$$

Square both side

$$\frac{V^2}{\omega^2} = A^2 - x^2$$

$$\frac{V^2}{\omega^2} + x^2 = A^2$$

divide by A^2 both side

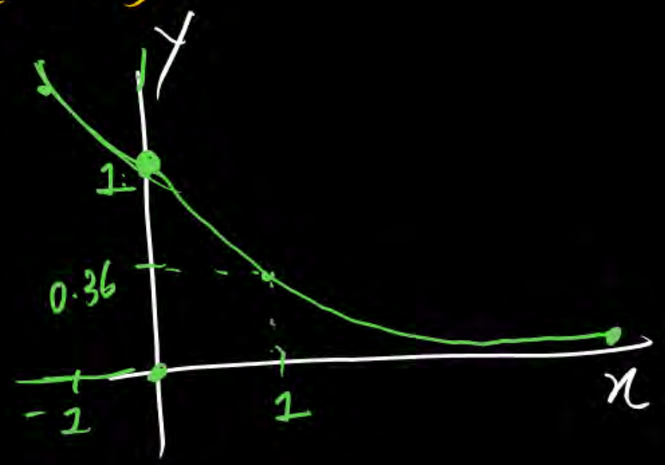
$$\frac{V^2}{(A\omega)^2} + \frac{x^2}{A^2} = 1$$

$$\frac{y^2}{(A\omega)^2} + \frac{x^2}{A^2} = 1$$

If $\omega = 1$ ⇐ graph will be circle

Exponential function (exponential decay graph)

$$y = e^{-x} = (2.71)^{-x}$$



Y	x
1	0
0.36	1
0	∞
+2.71	-1

$$y = e^{-1} = \frac{1}{e} = \frac{1}{2.71} = 0.36$$

$$y = e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$$

(Q) Find value of x so that Y will be negative $\boxed{Y = e^{-x}}$ → Not Possible

$x^0 = 1$ $\quad x \to$ non zero

$x^{\infty} = \infty$

$x^{-\infty} = \frac{1}{x^{\infty}} = \frac{1}{\infty} = 0$

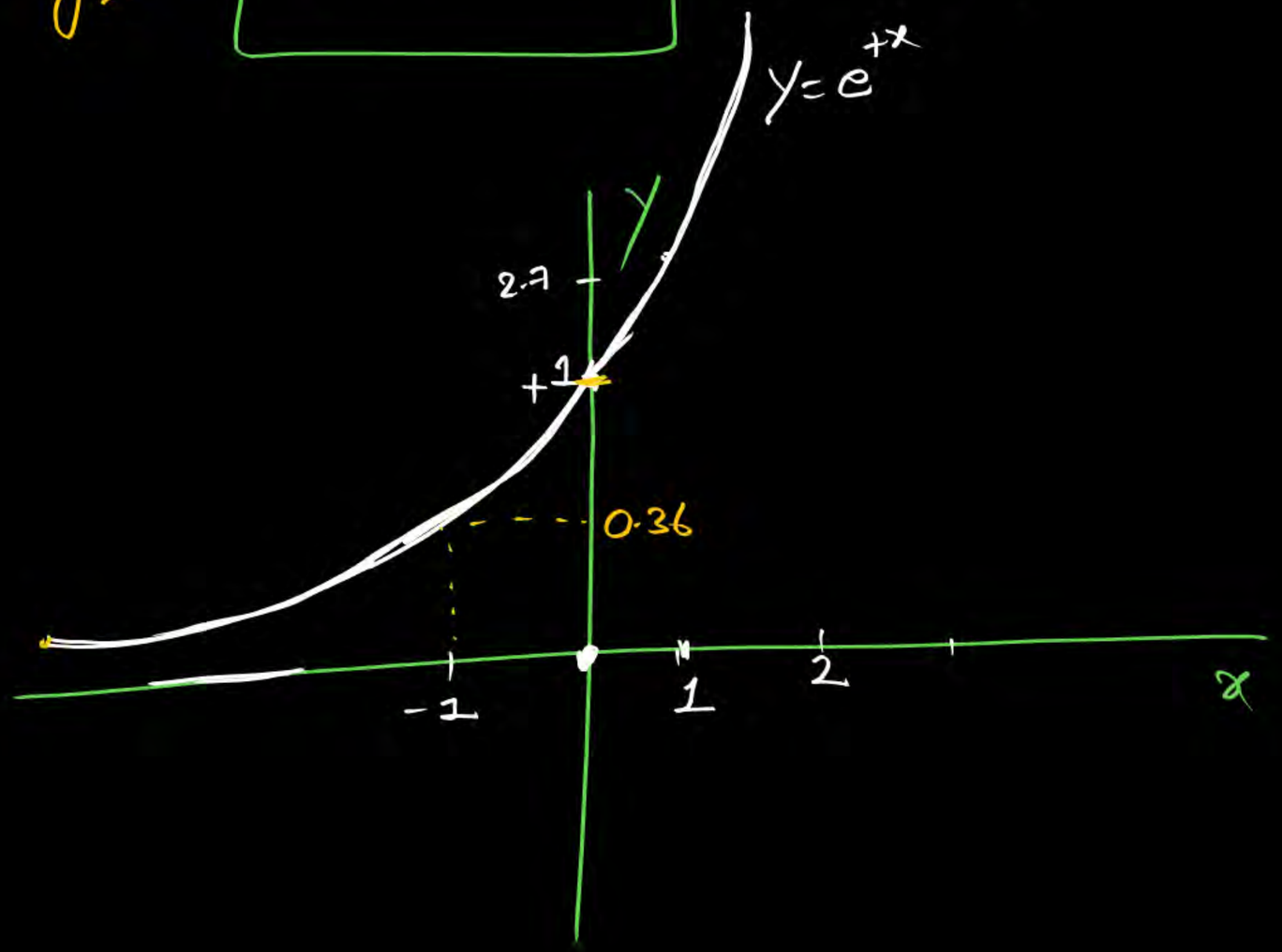
$\frac{1}{0} = \infty$ (infinity) $\quad \frac{1}{\infty} = 0$

$\frac{1}{x} = x^{-1}$

exponential growth

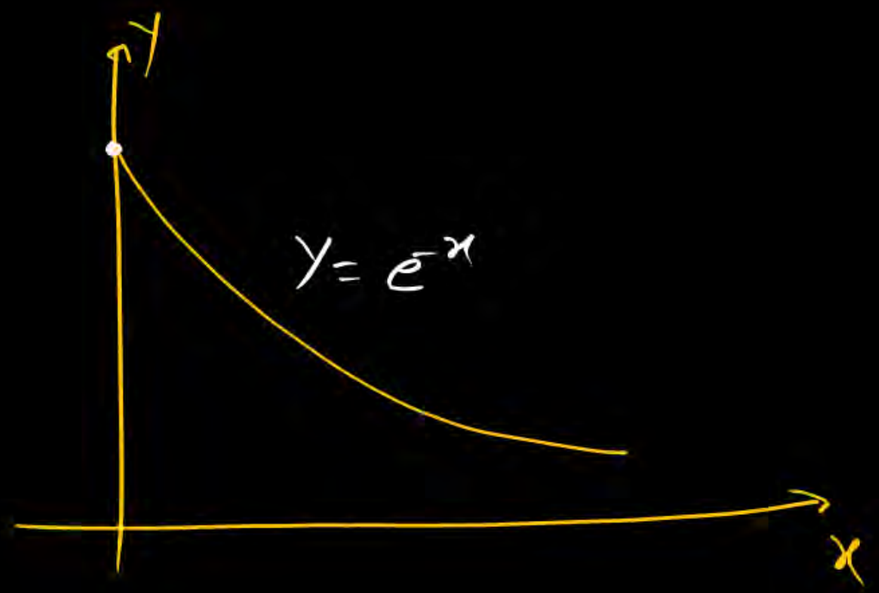
$$y = e^{x}$$

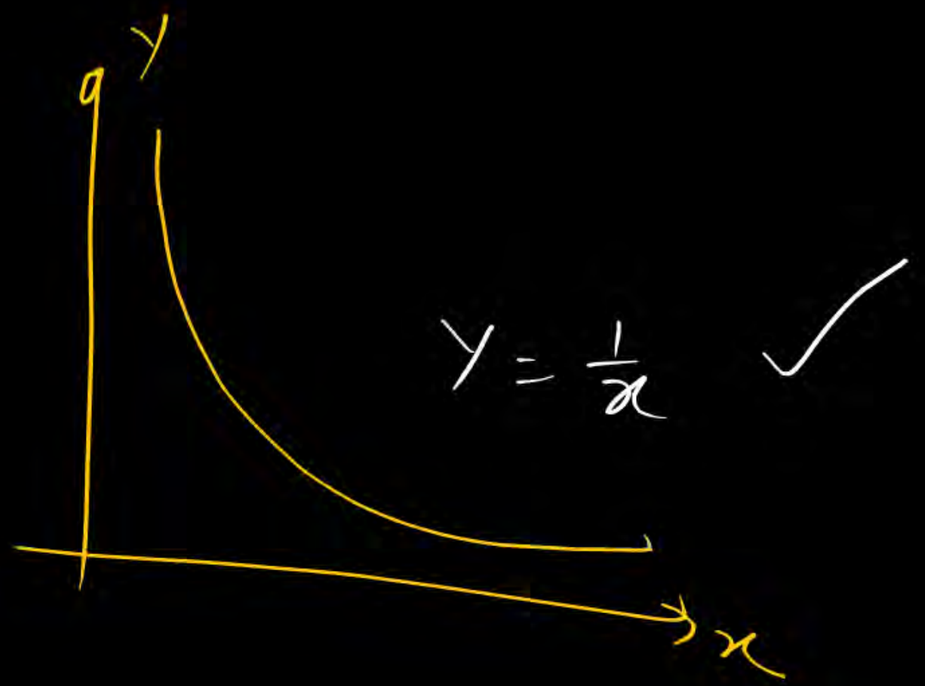
← Draw graph B/w y & x.



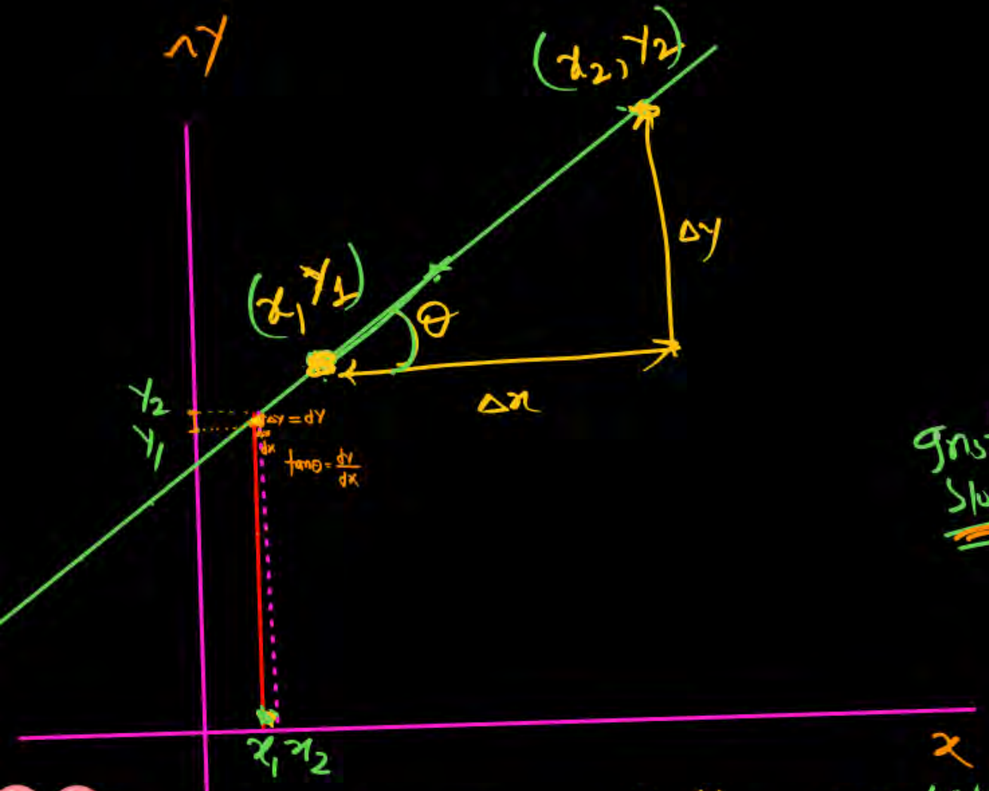
y	x
1	0
$e = 2.71$	1
$(2.71)^2$	2
0.36	-1
0	$-\infty$

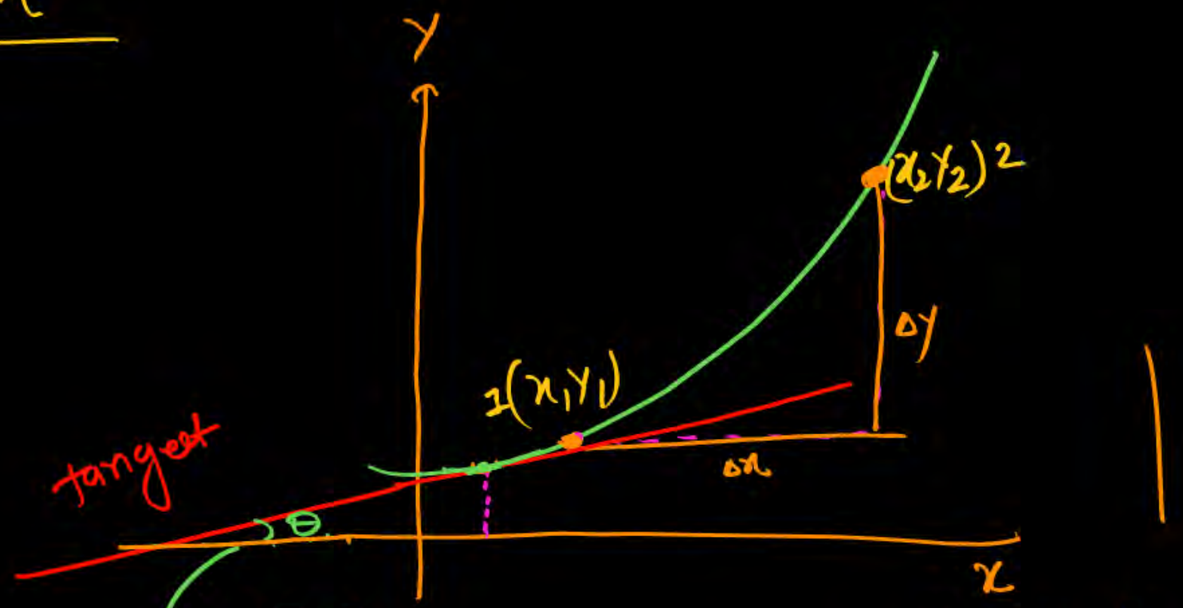
$$y = e^{x} = e^{-1} = \frac{1}{e} = 0.36$$


y
$y = e^{-x}$
x


y
$y = \frac{1}{x}$
x

Discussion on slope of all graph





inst slope $= \tan\theta$

Avg slope$_{12} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$ (No differentiation)

Instantaneous slope $= \frac{dy}{dx} = \frac{d(y)}{dx}$ = slope

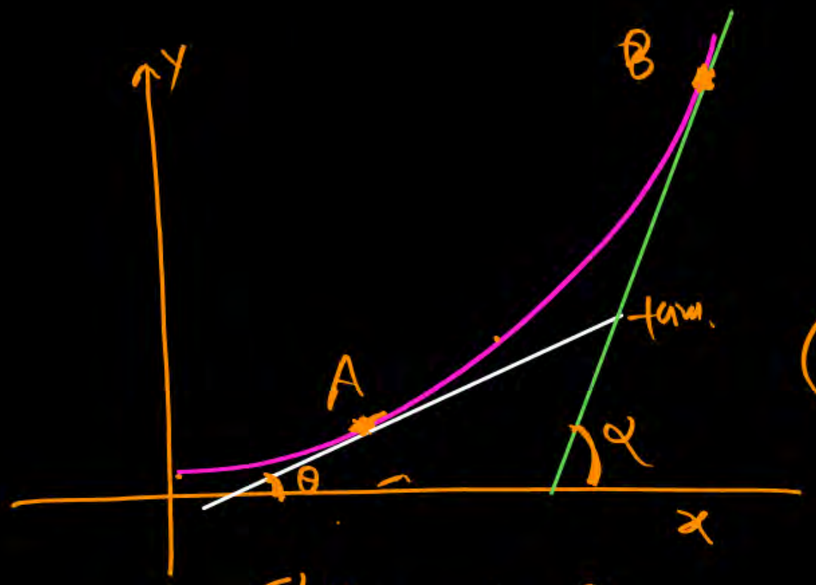
$\frac{dy}{dx}$ ← differentiation

$\tan\theta$ = Avg slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$ (No differentiation)

Instantaneous slope $= \frac{dy}{dx}$ ← differential of y w.r.t. x

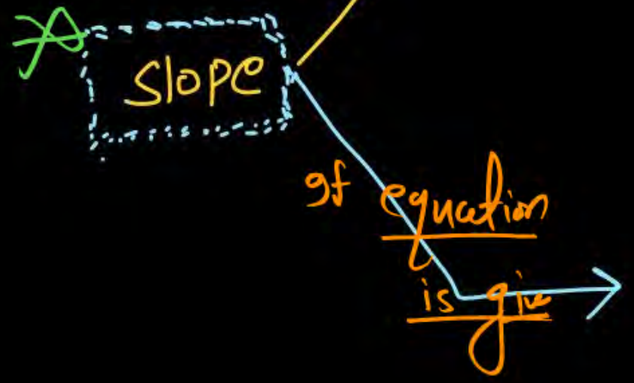
graph $\xrightarrow[\text{to MR}]{}$ chhu - kar mere tan ko; kiya Tune kya ishara ??

MR $\xrightarrow[\text{to graph}]{}$ chhu - kar tere tan ko kiya maine slope ka ishara.



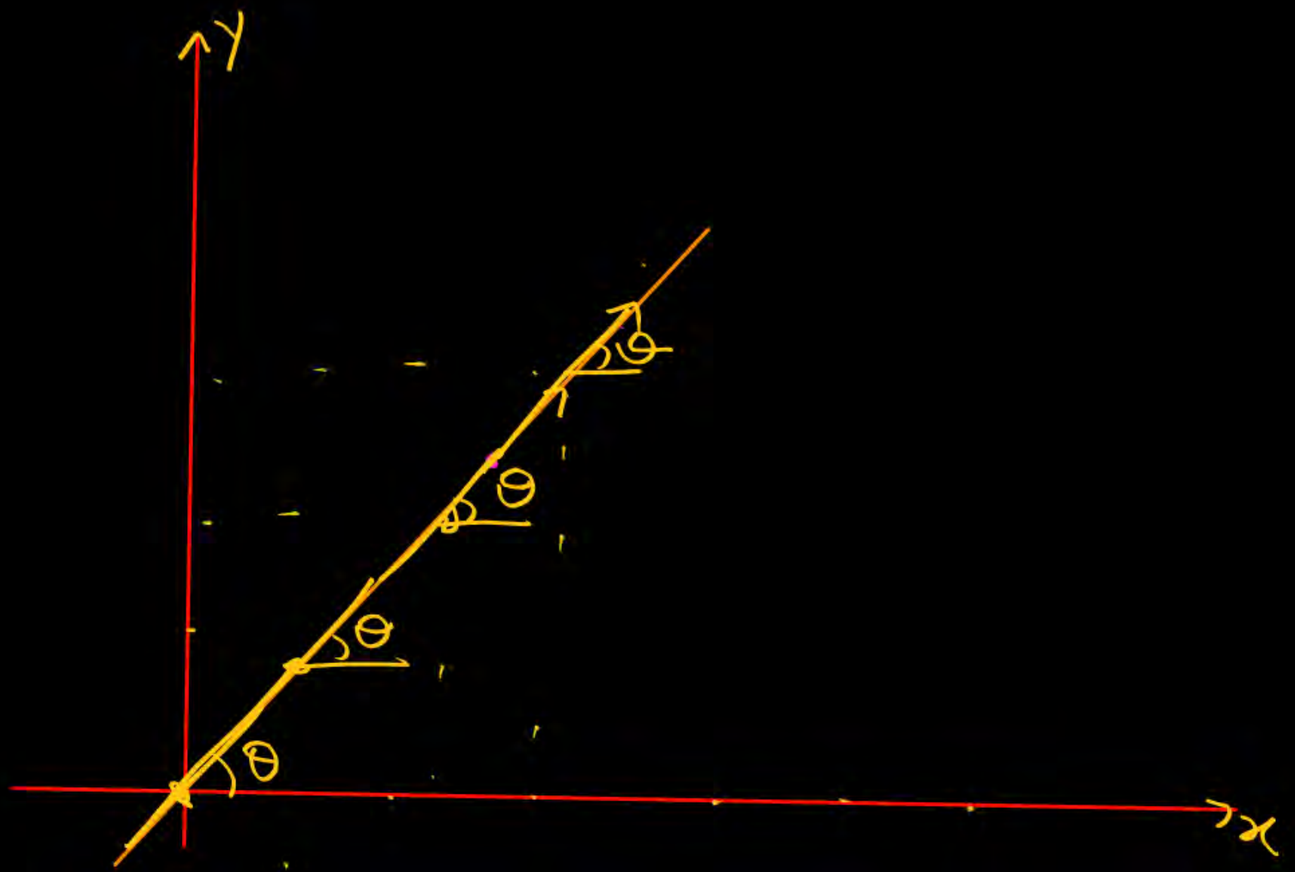
$(\text{slope})_B = \tan\alpha$

$\text{Slope}_A = \tan\theta$



Draw tangent at a point where slope have to find.
then slope = $\tan\theta$ where θ = Angle of tangent with x-axis.

slope at any point $\tan\theta = \frac{dy}{dx}$; diff^{n} of y w.r.t. x.



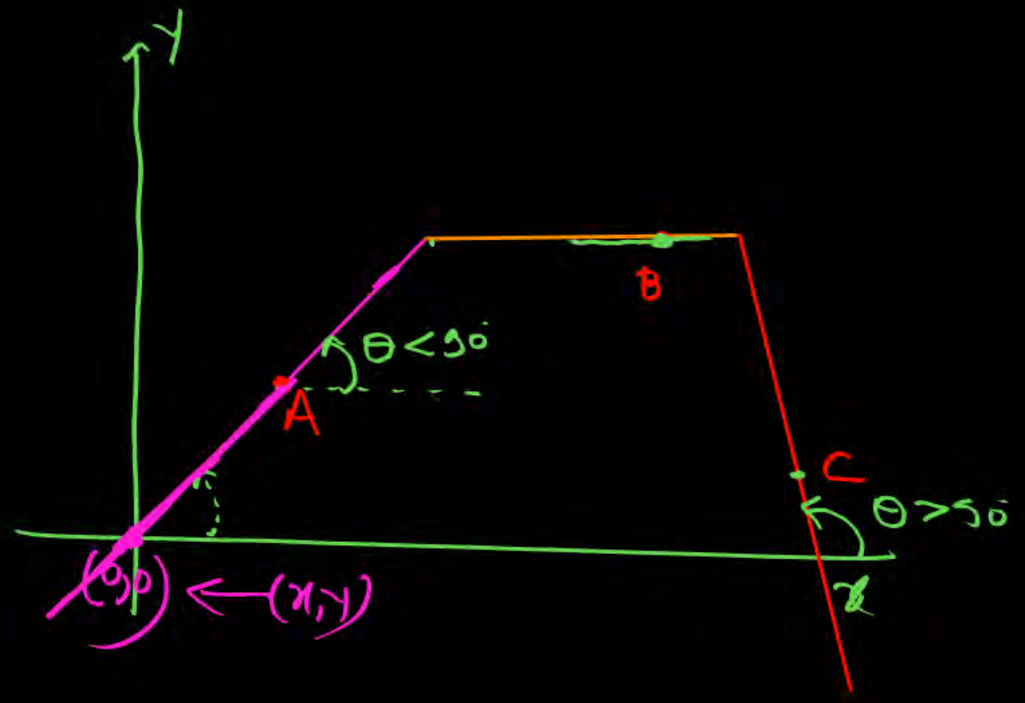
$x \uparrow$ and $y \uparrow$

Slope of a straight line
Same at all Point.

Slope = $\tan\theta$ ✓

↳ Same at point

①



At A

Slope_A = +ve

$\text{Slope}_B = 0$

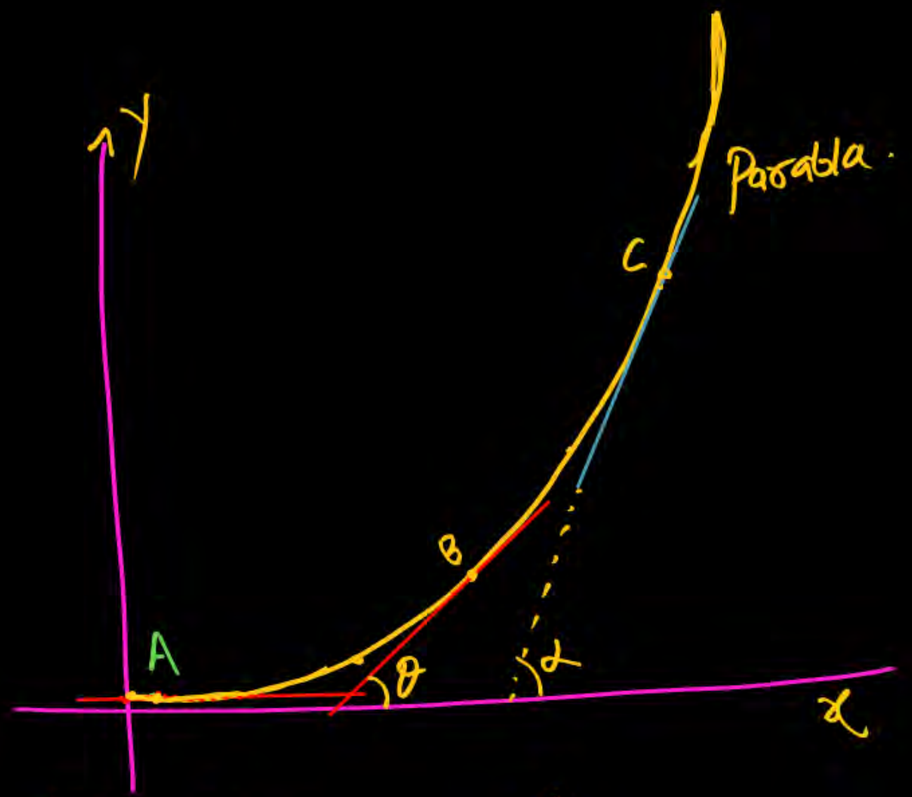
$(\text{Slope})_C$ = -ve

$\text{Slope} = \frac{y}{x}$ ✗

Slope at origin = (a) +ve (only 35%) ✓

(b) -ve ✗

(c) zero. 52%



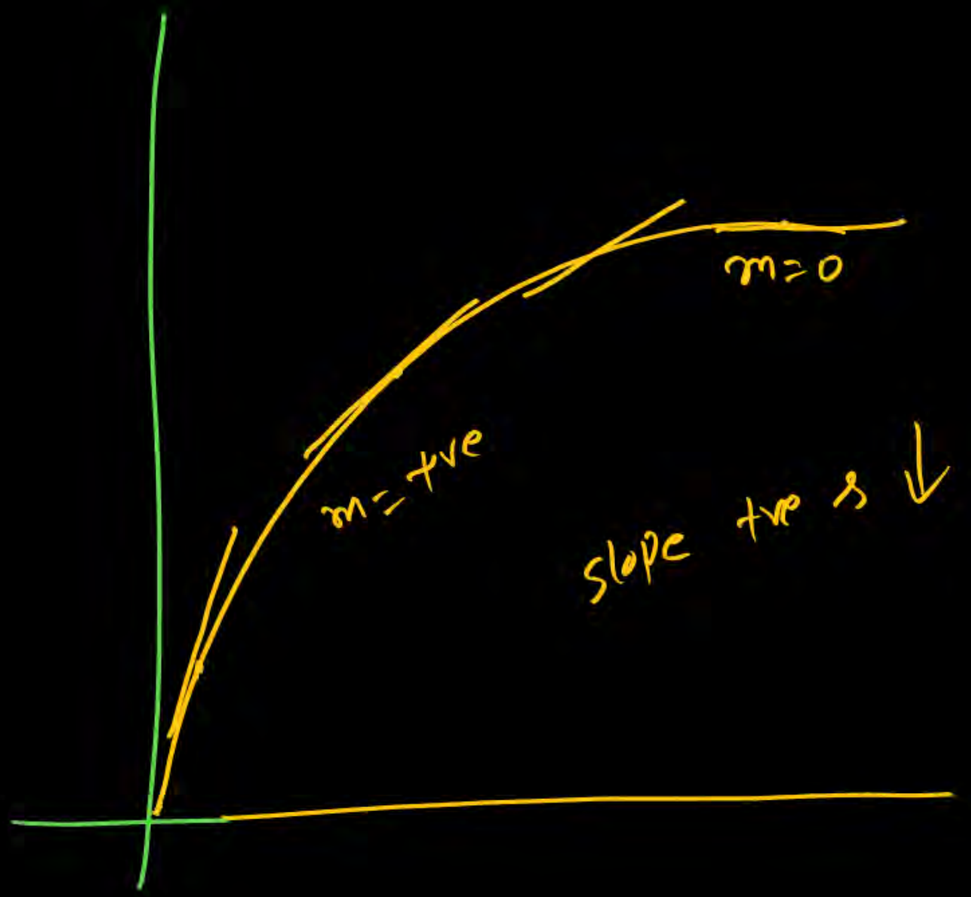
$\text{Slope}_A = \tan 0^\circ = 0$ ✓

Slope is +ve and Increasing

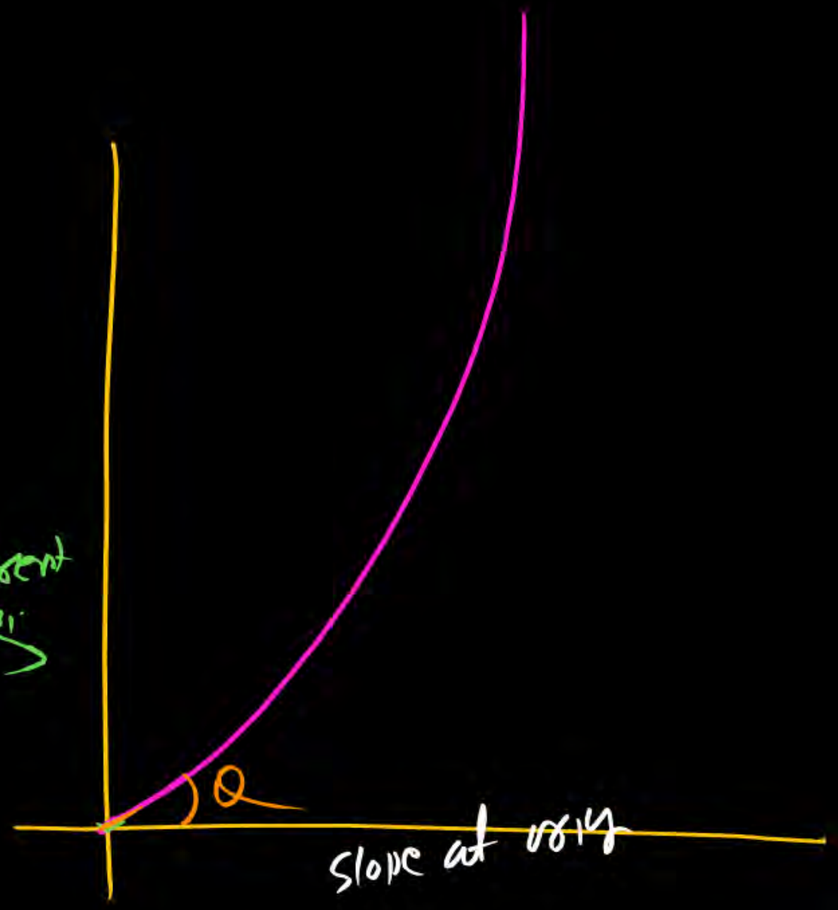
$\text{Slope}_B = +ve = \tan\theta$

$(\text{Slope})_C = +ve = \tan\alpha$

$\theta < \alpha$







(a) zero

(b) +ve

(c) -ve.

Copy

MR* Box

for slope



हसता हुआ Ramlal



रोता हुआ Ramlal

MR* Box

Magnitude of slope

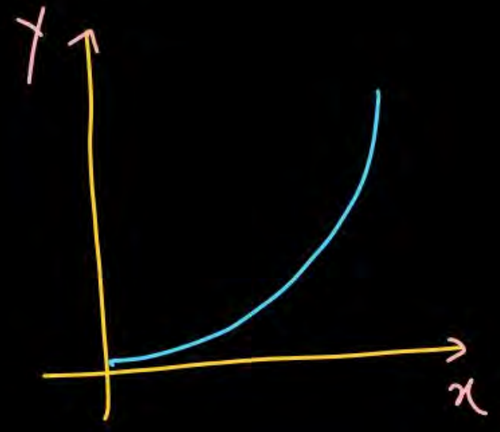
Horizontal & Vertical line

dekho → Horizontal Ki

Traf Jane se |m| ↓ decreasing

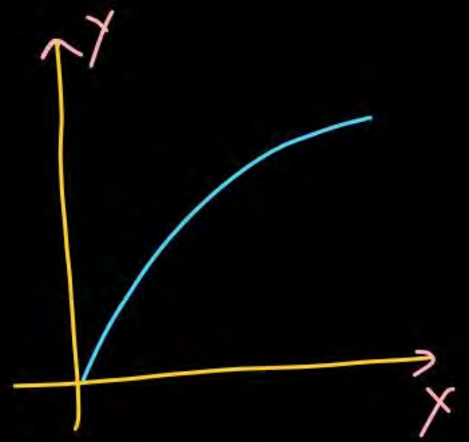
Horizontal se dur |m| ↑ increasing

Slope & magnitude of slope dono alag hota hai



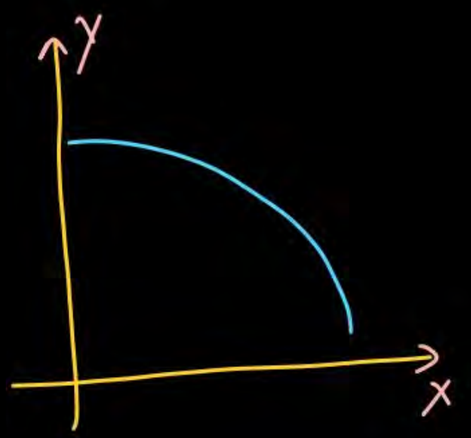
slope =

magnitude of slope =



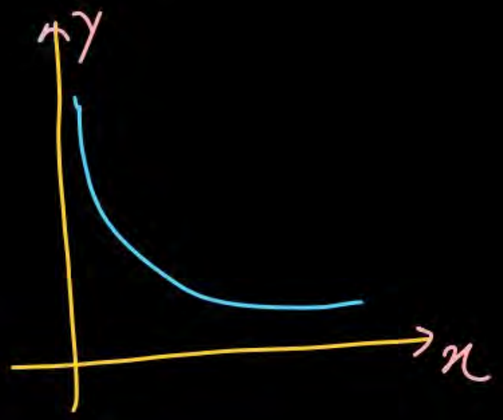
slope =

magnitude of slope =



slope =

magnitude of slope =



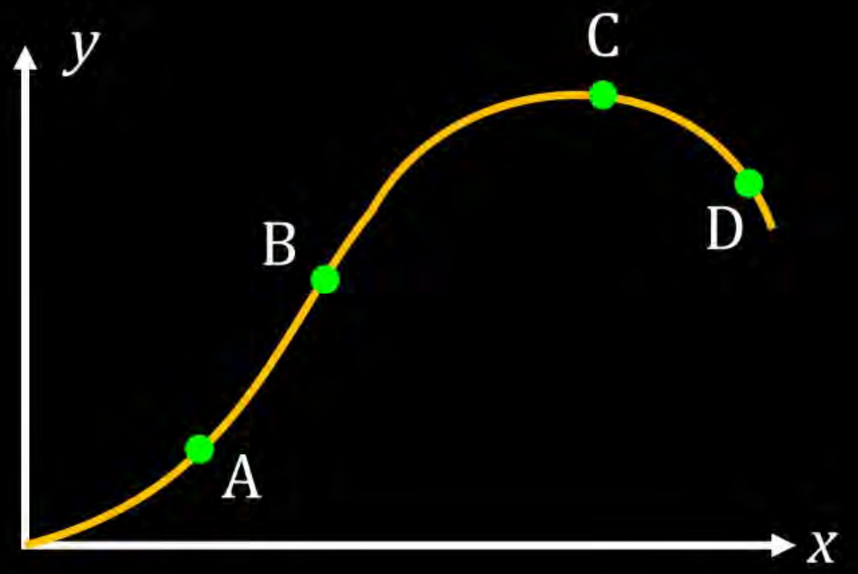
slope =

magnitude of slope =

Question

Match the matrix:

Point	Slope
A	Zero
B	negative
C	Maximum
D	Positive



Quadratic (10th class)

H/W



Question

$x^2 + 7x + 12 = 0$, find roots of 'x'.

Question

$2t^2 + 3t - 2 = 0$; find roots of equation.

Thank You